

TIN CITY, AK, USA

WMO: 701170

Lat: 65.563N

Lon: 167.921W

Elev: 83

StdP: 100.34

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26634

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-30.4	-28.5	-33.7	0.2	-29.9	-31.8	0.2	-28.4	23.2	-24.0	21.9	-21.0	13.0	10	1.178

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.1	16.4	12.7	14.2	11.8	12.7	10.9	13.4	15.6	12.3	14.0	11.1	12.5	4.7	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	9.1	14.2	11.3	8.4	13.0	10.2	7.8	11.7	37.9	15.6	34.9	14.1	32.1	12.4	16.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	19.1	17.5	DB	-32.3	20.3	2.6	2.2	-34.2	21.8	-35.8	23.1	-37.3	24.3	-39.2	25.9	(4)
(5)			WB	-31.8	15.3	3.0	1.0	-33.9	16.0	-35.7	16.6	-37.3	17.2	-39.5	17.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.3	-16.4	-16.1	-16.6	-9.6	-1.1	4.9	8.6	8.5	5.3	-0.3	-6.8	-13.1	(6)	
	DBStd	11.01	7.96	8.43	7.04	6.21	3.82	3.63	2.84	2.35	3.14	3.74	5.44	7.41	(7)	
	HDD10.0	5277	820	731	826	588	346	159	62	59	144	321	504	717	(8)	
	HDD18.3	8276	1078	964	1084	838	603	403	301	304	392	579	754	975	(9)	
	CDD10.0	42	0	0	0	0	1	7	19	13	2	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	9.0	11.2	11.7	10.8	10.1	8.0	7.3	6.7	7.2	7.3	7.7	8.5	10.9	(14)	
(15) Precipitation	PrecAvg	491	33	31	27	21	24	28	58	72	56	65	42	33	(15)	
	PrecMax	840	76	89	109	91	68	57	99	149	139	124	151	85	(16)	
	PrecMin	288	1	2	1	2	2	5	13	19	16	19	4	1	(17)	
	PrecStd	116	20	20	29	18	17	13	24	31	26	29	28	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	2.5	0.2	4.0	13.1	18.7	20.5	17.3	13.9	10.5	5.0	3.2	(19)	
		MCWB	1.4	1.5	-0.4	1.6	8.8	13.0	14.5	14.3	10.3	7.6	3.3	1.3	(20)	
	2%	DB	0.2	0.4	-1.0	2.2	7.9	15.1	17.0	14.7	12.0	7.8	2.9	0.9	(21)	
		MCWB	-0.4	0.0	-1.5	0.9	5.5	11.0	13.1	12.7	9.9	6.3	2.0	0.1	(22)	
	5%	DB	-1.3	-1.0	-2.3	0.8	5.3	12.5	14.8	13.2	10.6	5.7	1.7	-0.6	(23)	
		MCWB	-1.7	-1.3	-2.8	0.1	3.8	9.7	12.4	11.9	9.1	4.7	1.2	-1.2	(24)	
	10%	DB	-3.7	-2.5	-4.7	-0.7	3.1	10.4	12.9	12.0	9.3	4.1	0.4	-2.2	(25)	
		MCWB	-4.2	-2.9	-5.2	-1.0	1.9	8.5	11.2	11.0	8.3	3.3	0.0	-2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	1.4	0.0	2.3	9.1	13.4	15.1	14.6	11.5	7.9	3.9	1.5	(27)	
		MCDB	2.9	2.0	0.2	3.4	12.8	17.5	18.4	16.7	13.4	9.7	4.4	2.6	(28)	
	2%	WB	0.0	0.1	-1.2	0.9	6.0	11.5	13.8	13.1	10.2	6.3	2.2	0.1	(29)	
		MCDB	0.5	0.5	-1.0	1.7	8.0	14.6	16.3	14.2	11.4	7.8	2.6	0.8	(30)	
	5%	WB	-1.7	-1.1	-2.7	0.1	4.0	10.1	12.6	12.1	9.3	4.9	1.2	-1.1	(31)	
		MCDB	-1.4	-0.9	-2.4	0.8	5.3	12.3	14.7	13.1	10.4	5.8	1.6	-0.6	(32)	
	10%	WB	-4.2	-2.9	-5.4	-1.0	2.4	8.7	11.3	11.0	8.4	3.6	0.0	-2.6	(33)	
		MCDB	-3.7	-2.6	-4.8	-0.6	3.3	10.4	12.9	11.8	9.2	4.2	0.5	-2.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.6	5.2	5.1	3.8	4.7	4.1	3.6	3.3	3.3	4.3	5.0	(35)	
		MCDBR	7.4	6.9	6.0	5.3	7.1	8.4	8.0	5.7	4.4	3.7	3.8	6.0	(36)	
	5% WB	MCWBR	6.8	6.6	6.1	4.3	5.0	5.6	5.0	4.1	3.3	3.4	3.4	5.6	(37)	
		MCWBR	7.4	6.9	5.8	5.1	6.7	7.8	7.6	5.4	4.1	3.4	3.5	6.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.165	0.247	0.270	0.336	0.364	0.353	0.368	0.391	0.320	0.285	0.374	0.185	(40)	
		taud	1.902	2.086	2.153	1.991	2.133	2.429	2.434	2.304	2.492	2.280	2.701	1.899	(41)	
	Ebn at Noon		334	627	801	811	831	859	831	761	759	581	260	53	(42)	
		Edh at Noon	20	57	87	133	128	99	96	99	65	48	21	6	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.11	0.73	2.49	4.47	5.21	5.51	4.46	3.25	2.02	0.74	0.16	0.02	(44)	
	RadStd		0.01	0.09	0.20	0.33	0.54	0.47	0.41	0.36	0.19	0.07	0.02	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.83	N/A	N/A	N/A	+0.48	+0.55	N/A	N/A	N/A

Nomenclature: See separate page